

SplitStack Test Plan and Results Summary

Version: 1.0

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Related test case specification: `docs/splitstack_test_case_specification.md`

Test Strategy and Approach

SplitStack testing combines black-box functional testing, automated lint/build checks, and focused mobile widget/unit tests.

Primary strategy:

- Validate user-visible behavior against the main use cases.
- Confirm authentication prevents unauthenticated workspace access.
- Verify group and invite workflows across multiple users.
- Verify expense split validation and balance updates.
- Verify above-threshold expenses require voting before affecting balances.
- Verify settlements reduce open payment status.
- Verify receipt OCR extraction and receipt upload behavior.
- Verify analytics, budget, challenge, notification, and assistant flows.
- Run static checks and build checks before final submission.

Test Scope

In scope:

- Account registration and login.
- Group creation, editing, deletion, leave group, and invite response.
- Expense entry with equal, percent, custom, and personal split flows.
- Vote creation, approval, decline, and undo behavior.
- Dashboard balances and analytics.
- Settlement recording.
- Challenge creation and contribution.
- Settings and notification read state.
- AI assistant responses and confirmation-required actions.
- Web build/lint and backend lint.
- Existing Flutter tests for mobile UI and receipt parsing.

Out of scope for this class-project test plan:

- Production load testing over a multi-day window.
- Real payment processor integration testing.
- Public app-store release testing.

- Security penetration testing beyond membership and auth validation.

Test Environment

Item	Value
OS used for documentation validation	Windows, PowerShell
Node version observed	v22.20.0
Backend	Express API on port 3001
Web frontend	Vite dev server on port 5173
Database	Local SQLite database
Mobile	Flutter project under mobile-app/

Test Data

Recommended final walkthrough data:

- Register at least two real demo accounts.
- Create one shared group.
- Invite the second account by email.
- Add one below-threshold expense.
- Add one above-threshold expense to trigger a vote.
- Accept or decline the generated vote from all relevant members.
- Record one settlement payment.
- Add at least one personal expense.
- Add one challenge and one contribution.
- Upload at least one receipt image.
- Set a monthly budget with category breakdowns.
- Ask the assistant about balances and budget recommendations.

Entry Criteria

- Backend dependencies installed.
- Frontend dependencies installed.
- Backend starts successfully.
- Frontend starts successfully.
- Test users are available or can be registered.
- Mobile Flutter SDK is installed if mobile test execution is required.

Exit Criteria

- Critical user workflows pass in manual testing.
- Backend lint passes.
- Frontend lint passes.
- Frontend production build passes.
- Mobile test status is recorded.
- Known limitations are documented.

Traceability

Detailed test cases and traceability matrix are maintained in:

```
docs/splitstack_test_case_specification.md
```

That file maps test cases TC-01 through TC-16 to the main use cases UC-01 through UC-11.

Automated Validation Results

Commands run during this documentation pass:

Check	Command	Result	Notes
Backend lint	<code>npm.cmd run lint</code> from <code>website/backend</code>	Pass	ESLint completed successfully.
Frontend lint	<code>npm.cmd run lint</code> from <code>website/frontend</code>	Pass	ESLint completed successfully.
Frontend build	<code>npm.cmd run build</code> from <code>website/frontend</code>	Pass	Production build completed after resolving a local filesystem permission issue; output generated under <code>dist/</code> .
Flutter availability	<code>where.exe flutter</code>	Not available	Flutter CLI was not found in the current shell path.
Mobile tests	<code>flutter test</code> from <code>mobile-app</code>	Not run	Blocked because Flutter CLI was not available in this environment.

Existing Mobile Test Coverage

The repository includes Flutter tests in `mobile-app/test/` that cover:

- SplitStack login screen rendering.
- Receipt OCR amount extraction.
- Receipt OCR date extraction.

- Date picker editability after receipt autofill.
- Payout sheet payment method selection.
- Challenge tab rendering and contribution controls.

These should be run with:

```
cd mobile-app
flutter test
```

Manual Test Results Summary Template

Use this table during the final walkthrough or before submitting:

Test Case	Scenario	Result	Notes
TC-01	Register a new user account	Pending final walkthrough	
TC-02	Reject login with invalid credentials	Pending final walkthrough	
TC-03	Create a new group with invite	Pending final walkthrough	
TC-04	Accept a pending group invite	Pending final walkthrough	
TC-05	Add an equal split expense	Pending final walkthrough	
TC-06	Reject invalid percent split expense	Pending final walkthrough	
TC-07	Create vote for expense above threshold	Pending final walkthrough	
TC-08	Approve a pending expense vote	Pending final walkthrough	
TC-09	Decline a pending expense vote	Pending final walkthrough	
TC-10	View dashboard balances	Pending final walkthrough	
TC-11	Record a settlement payment	Pending final walkthrough	
TC-12	Create a savings challenge	Pending final walkthrough	
TC-13	Add contribution to savings challenge	Pending final walkthrough	
TC-14	View analytics	Pending final walkthrough	
TC-15	Update settings and mark notification read	Pending final walkthrough	
TC-16	Ask AI assistant a question	Pending final walkthrough	

Known Test Gaps

- Mobile tests were not executed in this environment because Flutter was unavailable.
- Manual end-to-end test results should be filled in after running the final demo flow.

- Performance and uptime criteria from the requirements document are specified but not fully measured in this local validation pass.
- Gemini-backed assistant behavior requires a valid `GEMINI_API_KEY` ; without it, local fallback behavior should be tested instead.